



University of Asia Pacific

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Third Year, Second Semester

Course Name: Software Engineering Lab

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Skillsphere

(Student Skill Tracker Platform)

1. Problem Statement

In today's competitive academic and professional environment, students are expected to develop a broad range of technical and soft skills beyond traditional classroom learning. While formal education provides essential foundations, employability increasingly depends on demonstrable skills, continuous learning, and practical experience. However, most students lack a structured and reliable system to track their skill development, learning progress, and achievements over time.

At present, student skills are scattered across resumes, online learning platforms, certificates, GitHub repositories, and personal records. This fragmentation makes it difficult to maintain a clear, consolidated, and verifiable skill profile. Consequently, students struggle to identify skill gaps, plan focused learning goals, and effectively present their competencies to instructors or potential employers.

Educational institutions and instructors face similar limitations. There is no unified platform to monitor, verify, and guide students' skill development beyond formal assessments. Instructors often rely on manual or informal evaluation methods, while recruiters and faculty lack visibility into students' practical abilities beyond academic transcripts.

Key Issues:

- **Lack of Centralized Skill Tracking:** Skills, projects, and certifications are distributed across multiple unconnected platforms.
- **Absence of Skill Verification:** There is no reliable system for instructors to validate claimed skills.
- **Limited Progress Monitoring:** Students cannot clearly measure or visualize their skill growth over time.
- **Inadequate Career Guidance:** Students lack structured skill roadmaps aligned with career paths.
- **Low Visibility of Student Potential:** Faculty and recruiters find it difficult to assess practical competencies.

- **Manual Evaluation Effort:** Instructors rely on time-consuming and unstructured evaluation processes.

These challenges clearly highlight the need for a comprehensive digital platform that systematically tracks, verifies, and visualizes student skills in a structured, transparent, and reliable manner.

2. Proposed Solution

The **Student Skill Tracker Platform** is a web-based system designed to provide a centralized, structured, and verifiable approach to skill development and monitoring. The platform empowers students to document, track, and showcase their skills while enabling instructors to validate progress and provide meaningful academic guidance.

Students can create personalized skill profiles, define proficiency levels, upload supporting evidence such as certificates or project links, and monitor their progress through visual indicators. Instructors can review submissions, verify skills, provide feedback, and recommend improvements. Administrators gain access to analytical insights that help evaluate skill trends across departments and programs.

Core Features:

- **User Authentication and Role-Based Access:** Secure access for students, instructors, and administrators.
- ☐ **Skill Profile Management:** Students can add skills, set proficiency levels, and update learning progress.
- ☐ **Evidence-Based Validation:** Upload certificates, project documentation, GitHub repositories, and external links.
- ☐ **Instructor Verification:** Faculty can approve, reject, or provide feedback on submitted skills.
- ☐ **Progress Visualization:** Skill-wise progress bars, timelines, and completion metrics.

☐ **Skill Roadmaps:** Predefined learning paths aligned with career goals such as Web Development or Data Science.

☐ **Achievement Badges:** Automated recognition based on verified skills and milestones.

User Experience Enhancements:

- **Mobile-Friendly Interface:** Accessible across smartphones, tablets, and desktops.
- **Personalized Dashboards:** Clear overview of progress, recommendations, and pending actions.
- **Notification System:** Alerts for feedback, verification updates, and achievements.
- **Portfolio Export:** Downloadable skill portfolio in PDF format for internships and job applications.
- **Analytics and Reports:** Department-wise and program-wise insights for administrators.

By bridging academic learning with practical skill development, the platform provides a scalable and transparent solution that benefits all stakeholders.

3. CEP (Complex Engineering Problem) Justification:

Complex Engineering knowledge

K3 – Depth of Engineering Knowledge

The system applies core software engineering principles, including system architecture design, database modeling, authentication, security mechanisms, and data processing techniques to ensure reliable and efficient operation.

K5 – Engineering Design and Problem Solving

The platform involves designing solutions to handle conflicting constraints such as user anonymity versus accountability, ease of use versus data security, and flexibility versus system control.

K6 – Engineering Practice and Tools

The system utilizes modern engineering tools and practices, including web development frameworks, database management systems, cloud or server deployment, and analytical tools for data visualization and reporting.

K7 – Impact, Ethics, and Responsibility

The platform considers ethical and societal responsibilities by ensuring data privacy, secure handling of student feedback, and meaningful impact on teaching quality and institutional decision-making.

Complex Engineering Problem

P1 – In-depth Engineering Knowledge

The system requires strong foundations in software engineering, database management, authentication mechanisms, file handling, and data visualization.

P2 – Conflicting Requirements

The platform balances usability with security, student flexibility with instructor control, and scalability with performance.

P6 – Multiple Stakeholders

The system serves students, instructors, administrators, and potentially recruiters, each with distinct requirements and expectations.

Complex Engineering Problem A1 –

Use of Diverse Resources

The system integrates human users, databases, cloud storage, servers, and analytical tools.

A2 – Interaction of Multiple Issues

It simultaneously addresses usability, security, verification workflows, performance optimization, and data integrity.

A3 – Innovation

The platform uniquely combines skill tracking, evidence-based verification, and progress visualization in a unified system.

A4 – Impact on Education and Society

It enhances employability, supports outcome-based education, and promotes continuous learning.

A5 – Beyond Previous Experience

The project moves beyond traditional grading systems toward competency- based skill evaluation.

4. Similar Applications:

Several existing platforms provide partial solutions related to skill development:

- **LinkedIn Skill Profiles:** Allow skill listing and endorsements but lack academic validation and progress tracking.
- **Coursera and Udemy:** Provide course certificates but do not unify skills across platforms.
- **GitHub:** Showcases coding projects but lacks structured evaluation and instructor feedback.
- **University LMS (e.g., Moodle):** Focus on course delivery and grades rather than long-term skill progression.

Unlike these platforms, the Student Skill Tracker Platform integrates skill tracking, verification, analytics, and academic guidance into a single cohesive system.

Questionnaires for Survey and Interview:

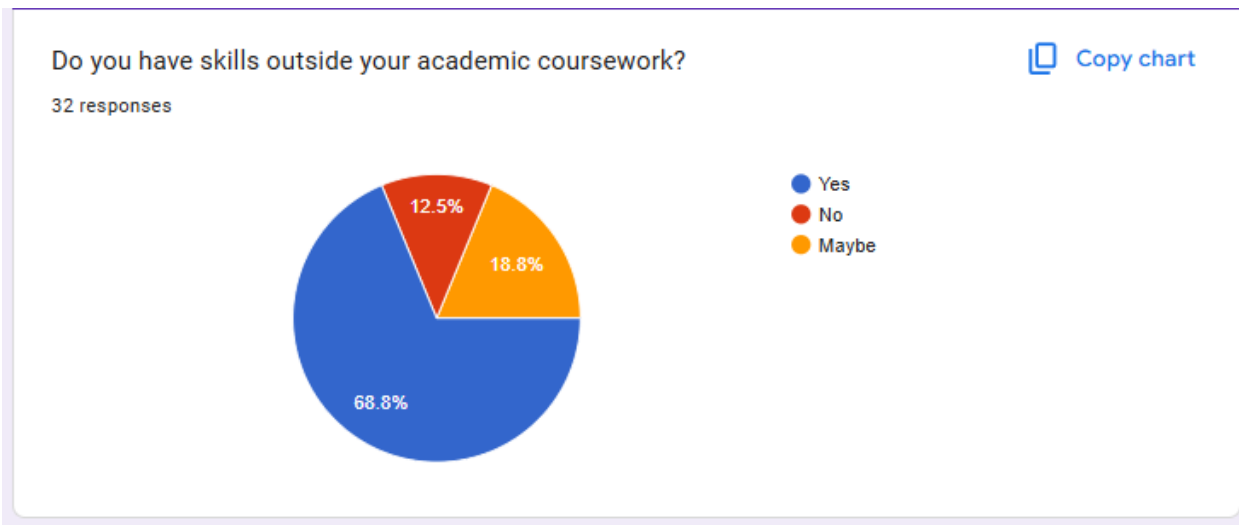
For Survey

1. Your Name?
2. Your university name?
3. Your department/major?
4. Your CGPA range?
5. Do you have skills outside your academic coursework?
6. Which types of skills do you have? (Select all that apply)
7. Have you completed any of the following?
8. Do you maintain a portfolio (GitHub, Drive, etc.)?
9. Have you applied for internships or part-time jobs?
10. What is your biggest challenge in getting internships/jobs?
11. Do you feel your CV properly represents your actual skills?
12. Do you think recruiters trust student CV skills?
13. Would you like your skills to be officially verified by a teacher?
14. Do you believe verified skills would increase your chance of getting selected?
15. If there was a university-based student skill platform (like a student-only job portal), would you use it?
16. What features would you prefer? (Select all that apply)
17. Would you recommend such a platform to your friends?
18. What improvement do you want in current job platforms for students?
19. Do you think there is a lack of a proper platform dedicated only to students for internships and skill showcasing?
20. Do you feel general job portals are not fully suitable for students?
21. Have you ever felt your skills were not properly recognized when applying for jobs/internships?

22. Do you think many students exaggerate skills in their CV?
23. Would you create a profile on such a platform?
24. Do you think this type of platform is necessary?
25. Would you request your teacher to verify your skills?
26. Do you believe teacher verification would increase recruiter trust?
27. Which features are most important to you?
28. Would you be willing to pay a small amount for premium features (e.g., better visibility)?
29. Do you think this platform could improve student employability?
30. Do you think this system would reduce fake skill claims?
31. Would you recommend this platform to your university?
32. What problem do you face in showcasing your skills?
33. What additional feature should be added to this platform?
34. Do you have any concerns about this system?

REQUIREMENT ANALYSIS

Survey Analysis Report:

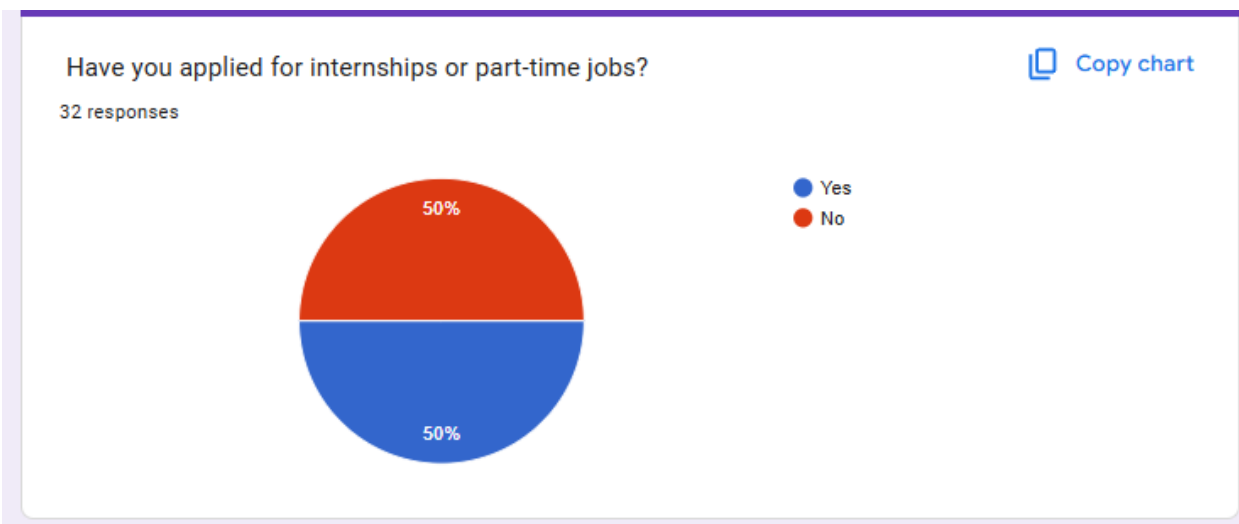


Result:

68.8 % answered Yes.
18.8 % answered Maybe.
12.5 % answered No.

Insight:

The majority of students possess non-academic or extra skills. This shows students already have diverse competencies, but these skills need structured documentation and visibility.



Result:

50 % applied for internships.
50 % did not.

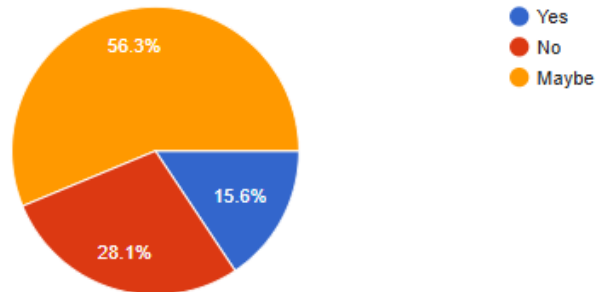
Insight:

There is moderate engagement in job seeking, but barriers may exist.

Do you feel your CV properly represents your actual skills?

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32 responses



Result:

56.3 % answered Maybe.

28.1 % answered No.

Only 15.6 % answered Yes.

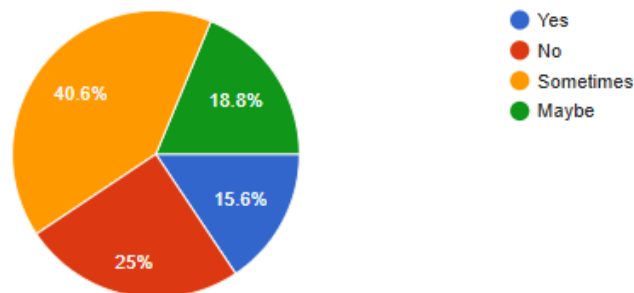
Insight:

Most students lack confidence that their CV truly represents their abilities. This indicates a credibility gap.

Do you think recruiters trust student CV skills?

 [Copy chart](#)

32 responses



Result:

40.6 % said Sometimes.

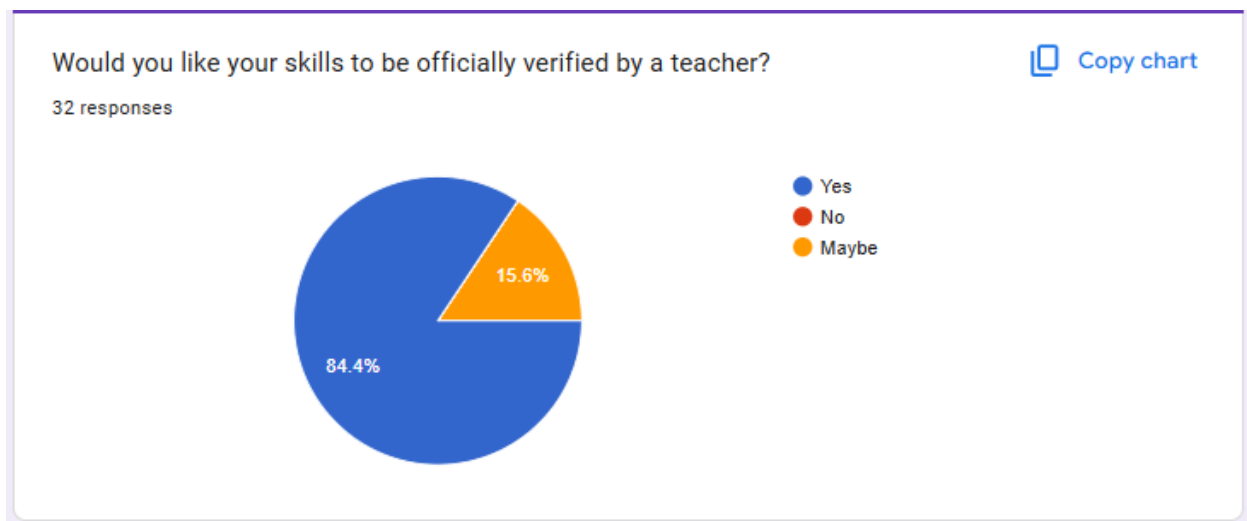
25 % said No.

18.8 % said Maybe.

15.6 % said Yes.

Insight:

There is strong uncertainty about recruiter trust. Verification can reduce this trust gap



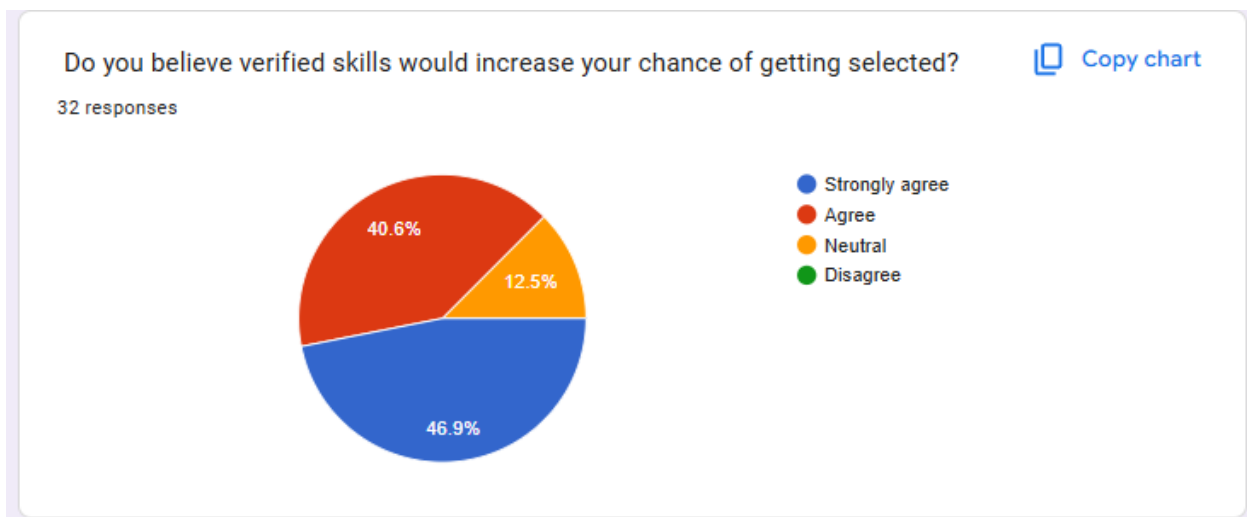
Result:

84.4 % said Yes.

15.6 % said Maybe.

Insight:

There is overwhelming support for teacher verification, validating your core system concept.



Result:

46.9 % strongly agree.

40.6 % agree.

Only 12.5 % neutral.

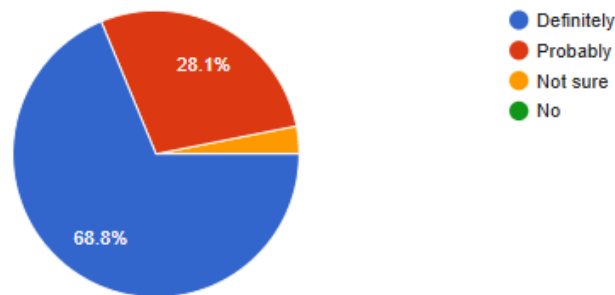
Insight:

Nearly 87 % believe verification improves employability. Strong validation of the idea.

If there was a university-based student skill platform (like a student-only job portal), would you use it?

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32 responses



Result:

68.8 % Definitely.

28.1 % Probably.

Very few unsure.

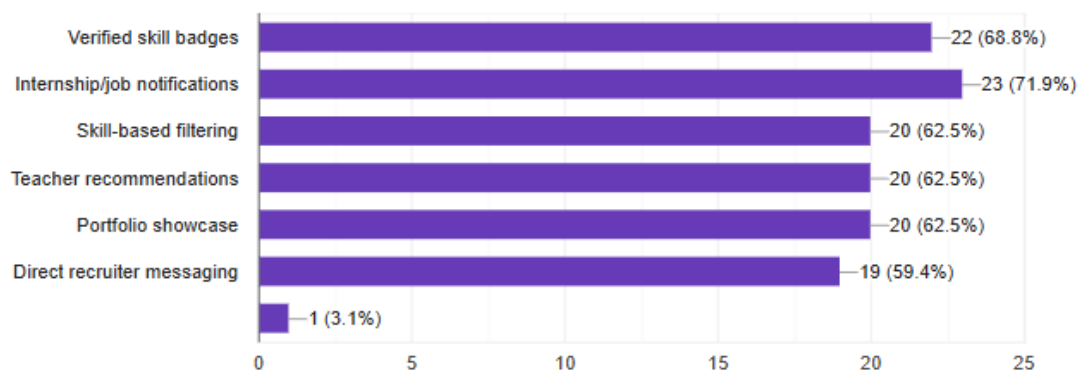
Insight:

High adoption potential. Students are ready to use such a system.

What features would you prefer? (Select all that apply)

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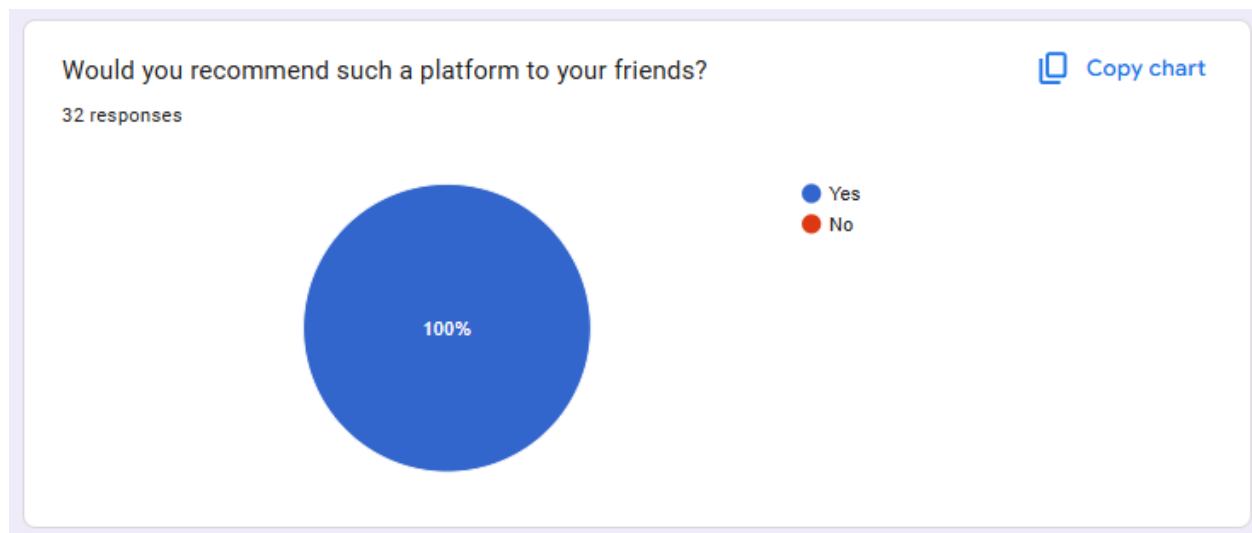
32 responses



- Internship notifications (71.9 %)
- Verified skill badges (68.8 %)
- Skill-based filtering (62.5 %)
- Portfolio showcase (62.5 %)

Insight:

Core modules should include verification, matching, and filtering systems

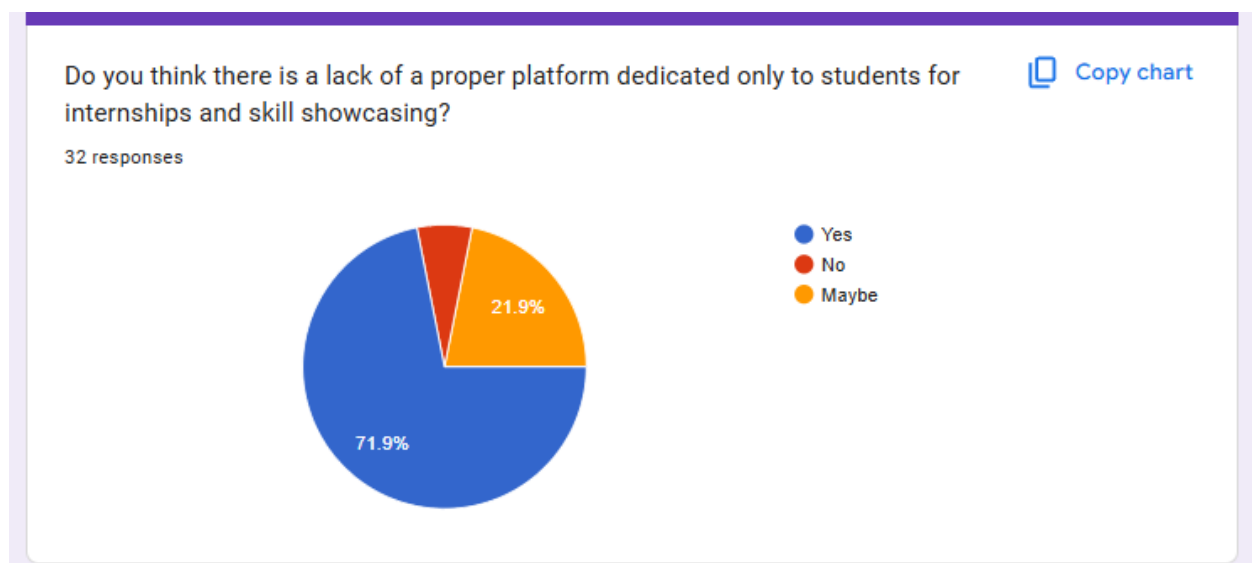


Result:

100 % Yes.

Insight:

Strong social acceptance and potential organic growth.



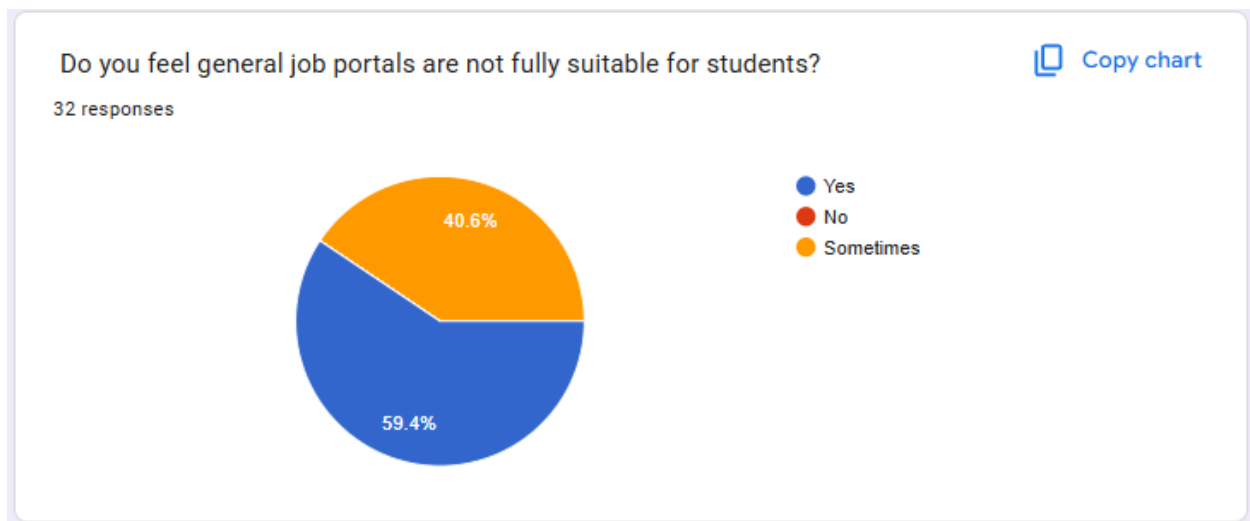
Result:

71.9 % Yes.

21.9 % Maybe.

Insight:

Clear perceived gap in the current market.



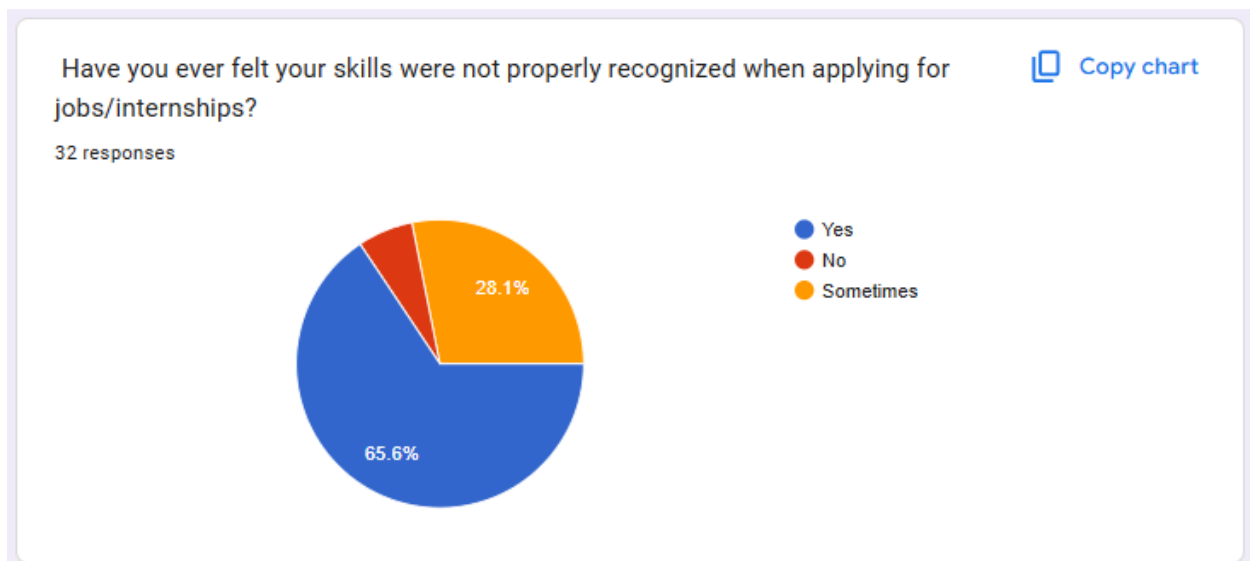
Result:

59.4 % No.

40.6 % Sometimes.

Insight:

Students feel general job portals are not fully tailored for them.



Result:

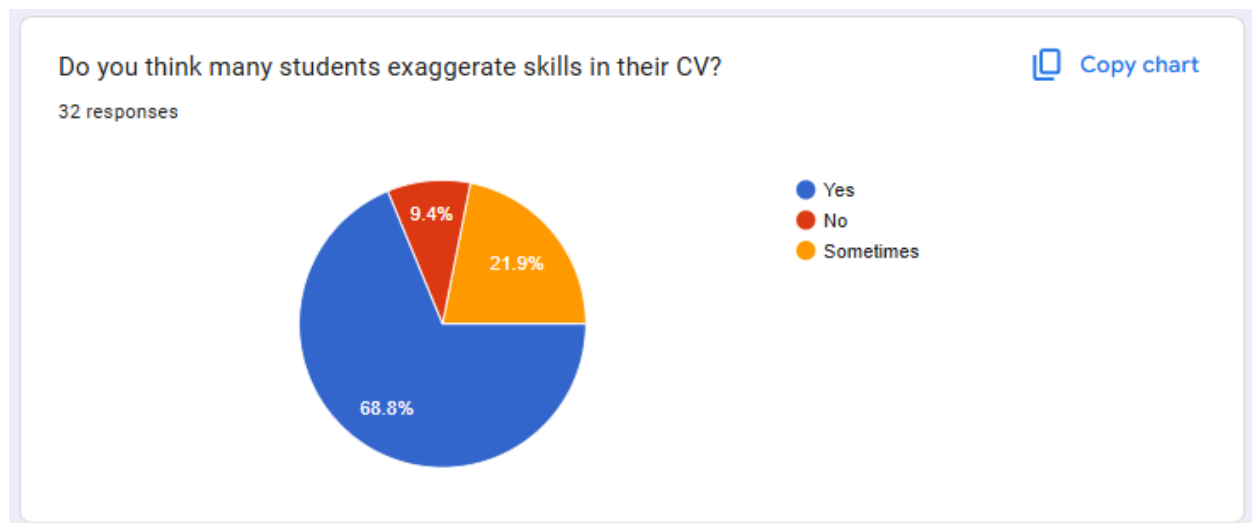
65.6 % Yes.

6.3 % said No.

28.1 % said Maybe.

Insight:

Majority feel undervalued. This supports the need for verified recognition.



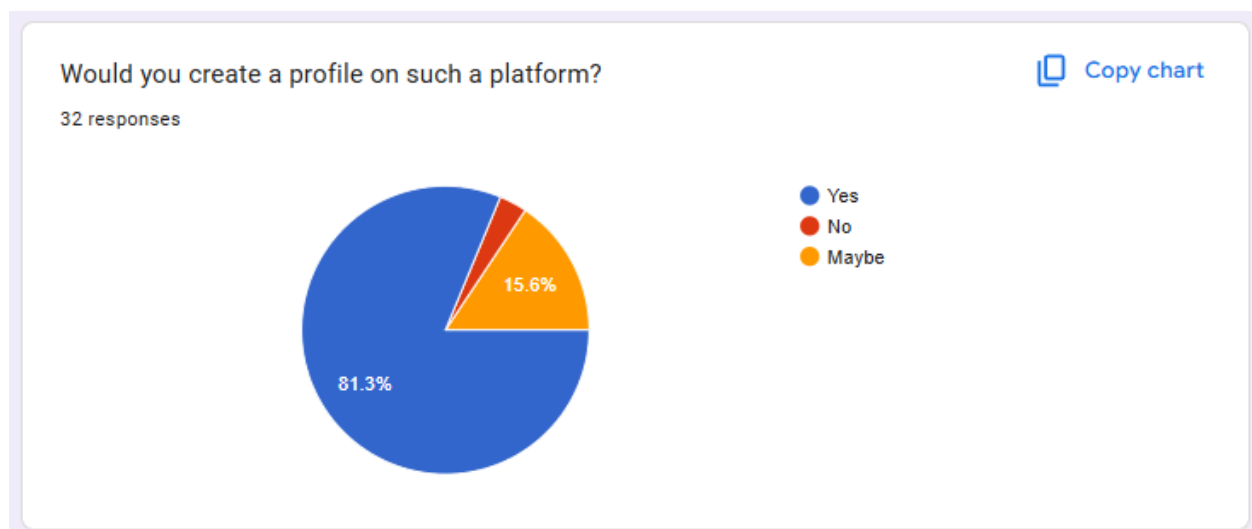
Result:

68.8 % Yes.

21.9 % Sometimes.

Insight:

There is awareness of exaggeration. Verification can reduce false claims.



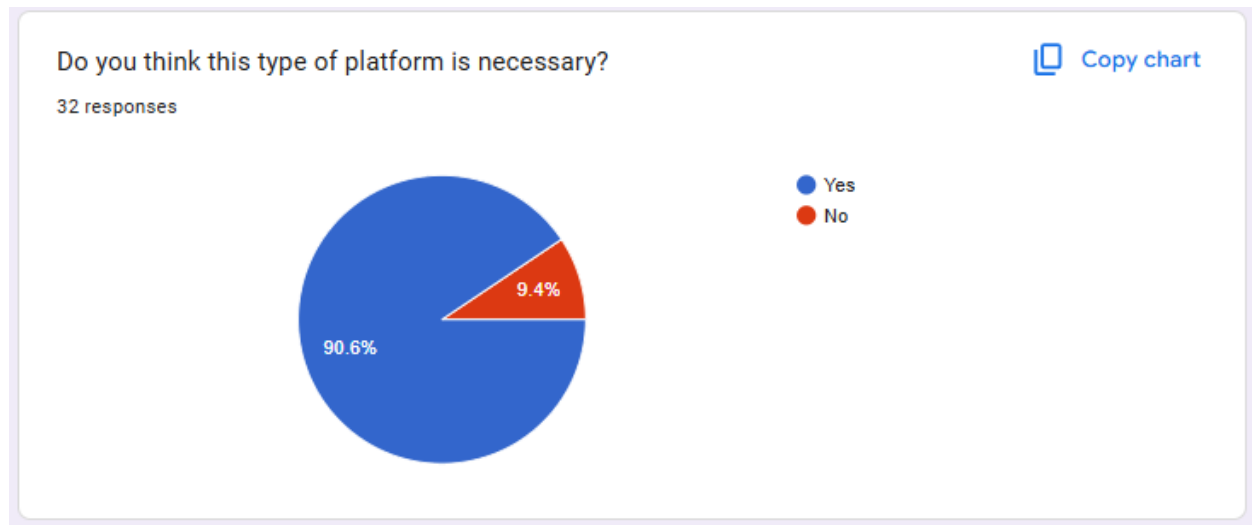
Result:

81.3 % Yes.

15.6 % Maybe.

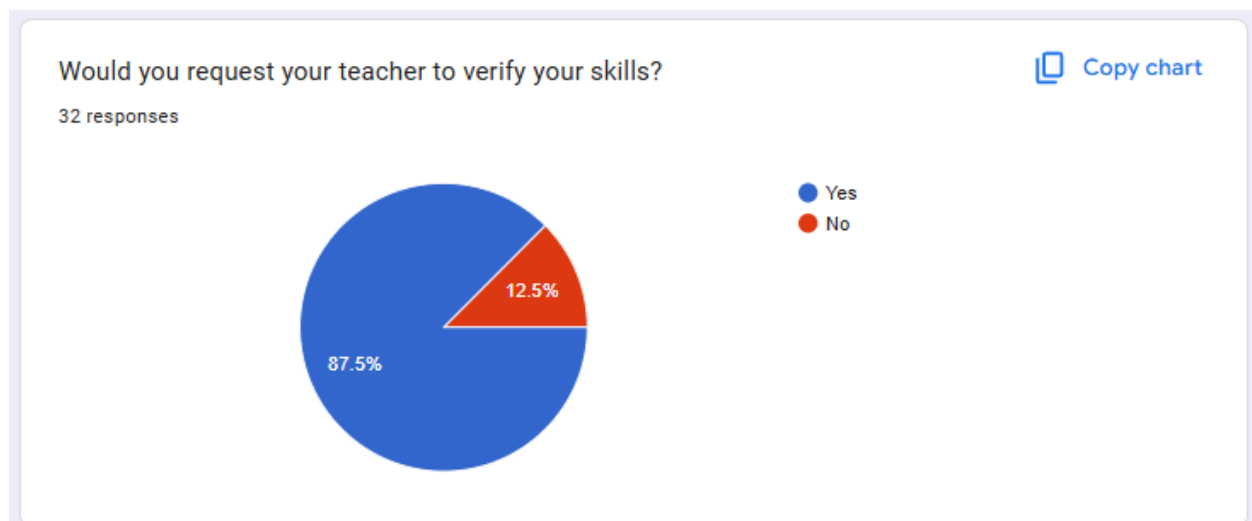
Insight:

Strong willingness to adopt.



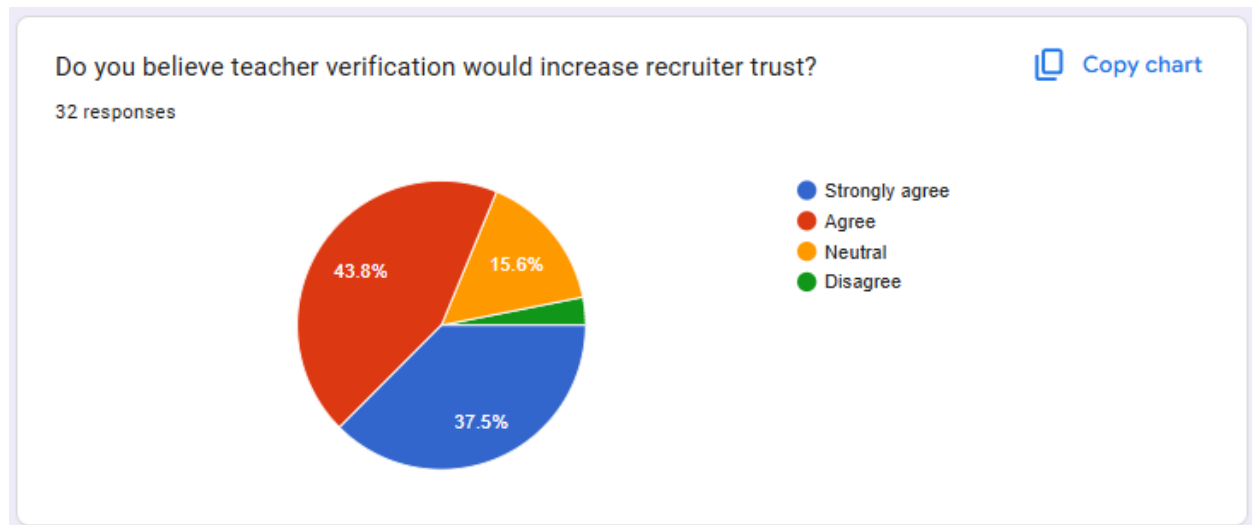
Result:
90.6 % Yes.

Insight:
Overwhelming support for system necessity.



Result:
87.5 % Yes.

Insight:
Students are ready to engage with teachers in verification process.



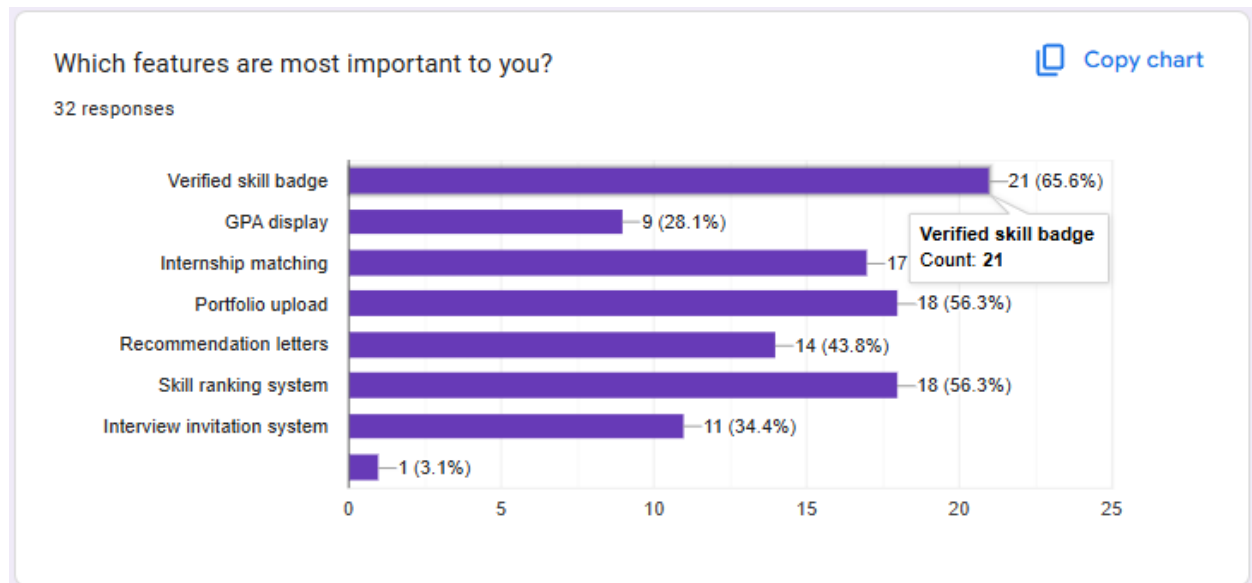
Result:

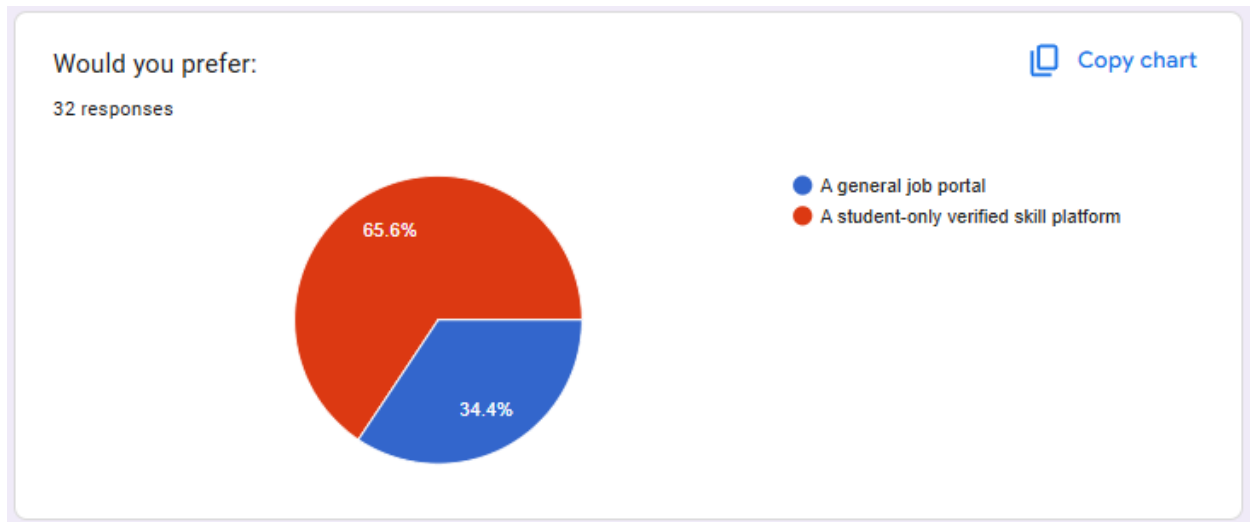
37.5 % strongly agree.

43.8 % agree.

Insight:

More than 80% believe verification increases trust.



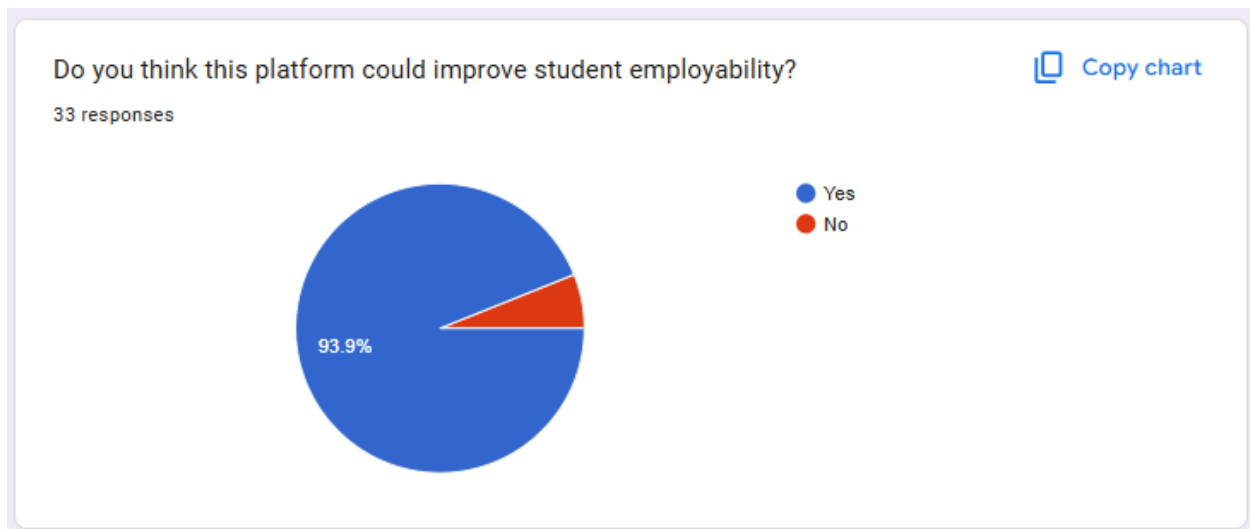


Result:

65.6 % prefer student-only verified platform over general job portal.

Insight:

Clear competitive advantage over traditional job portals.

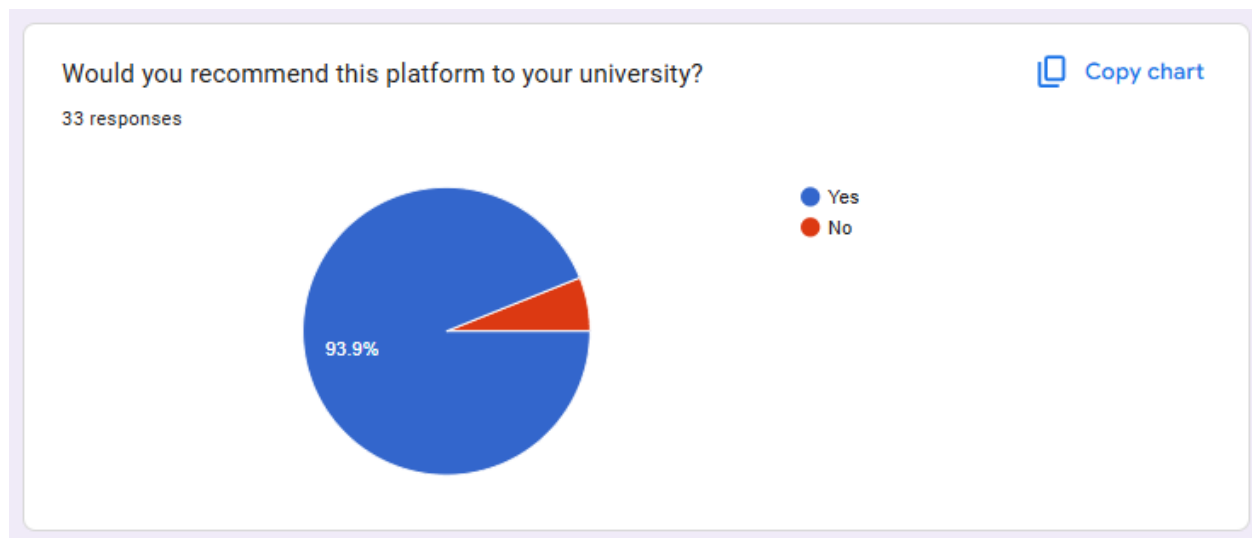


Result

93.9 % Yes.

Insight:

Strong perceived impact on career success.

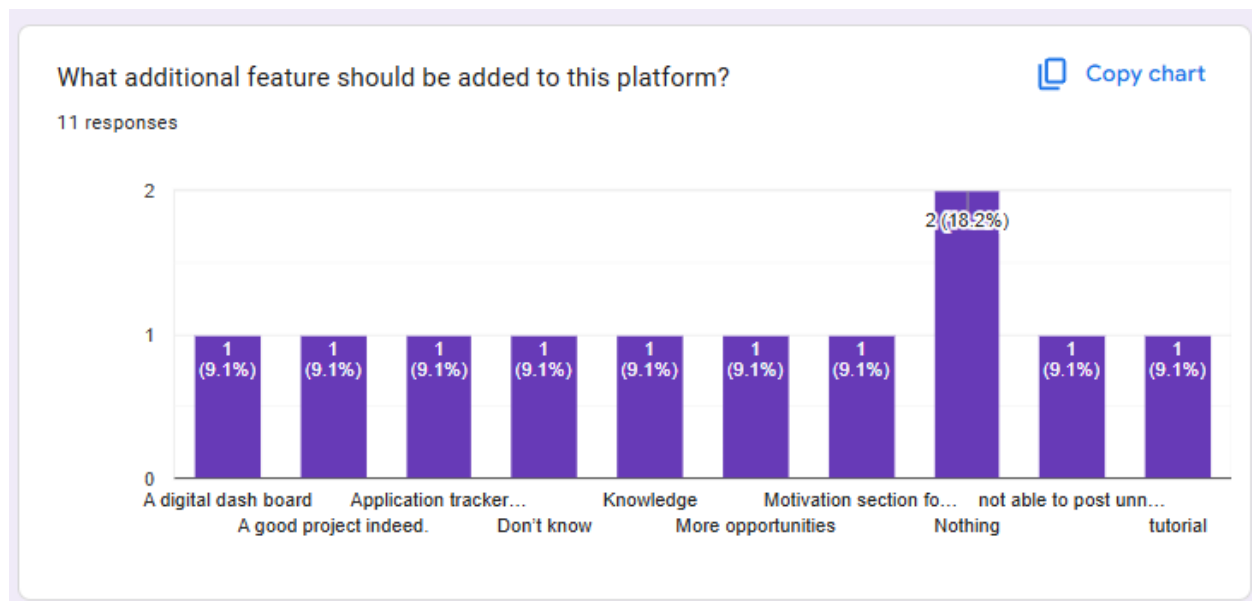


Result:

93.9 % Yes.

Insight:

Institutional adoption potential is high.



For Interview

Questions for Faculty

1. What problems do you notice in how students show their skills on CVs or job websites?
2. Do you think students sometimes overstate their skills? Why?
3. How do you usually check or evaluate a student's practical or technical skills?
4. Do you think teachers should help verify student skills for jobs? Why or why not?
5. What should be considered when verifying a student's skills?
6. Would you be willing to verify student skills through a simple online system?
7. Do you have any concerns about verifying student skills on a digital platform?
8. Do you think verified skill badges would help recruiters trust students more?
9. How could this platform help connect universities with companies?
10. What possible problems or risks should be considered in this system?
11. How can we make sure the verification process is fair for all students?
12. What features would you suggest to improve this platform?

Questions for Student

1. Do you think there is a proper platform only for students to showcase their skills?
2. What problems do you face when applying for internships or jobs?
3. Do you think your CV properly shows your real skills?
4. Have you ever felt that recruiters did not trust your listed skills?
5. Do you think some students exaggerate their skills on CVs?
6. Would you like your skills to be officially verified by a teacher? Why or why not?
7. Do you think verified skills would increase your chances of getting selected?
8. Would you use a student-only skill platform for internships and jobs?
9. What features would you want in such a platform?
10. Do you think this type of platform could improve student employability?
11. What concerns would you have about this system?
12. What suggestions do you have to improve this idea?

Questions for Recruiters

1. What challenges do you face when hiring students or fresh graduates?
2. Do you find it difficult to verify the skills mentioned in student CVs?
3. How do you currently assess whether a student actually has the skills they claim?
4. Would you prefer candidates with verified skills from university teachers? Why?
5. Do you think a student-only recruitment platform would be useful for your organization?
6. What information is most important to you when shortlisting student candidates?
7. Would skill-based filtering (e.g., verified programming, design, communication skills) help you hire faster?
8. Do you think a verified badge system would increase your trust in applicants?
9. What concerns would you have about using a university-based student recruitment platform?
10. What features would you suggest to make such a platform valuable for recruiters?

THE INTERVIEW:

Faculties:

Interviewee 01:

Name : Baivab Das

Occupation: Assistant Professor, Dept. of CSE, University of Asia Pacific

Interviewer: MD. Fairuz Anam

I chose him for my interview because of his strong academic background. His over 3 years of university teaching experience make him highly knowledgeable and experienced. His work in advanced research areas and commitment to education make him an inspiring and suitable person to interview.

1. What problems do you notice in how students show their skills on CVs or job websites?

- ❖ One common issue I notice is that students often list skills without explaining their level of proficiency or if they have actually done those practically or not. Many CVs look similar, with long lists of tools and technologies, but there is little evidence to show depth of understanding or real application.

2. Do you think students sometimes overstate their skills? Why?

- ❖ Yes, it does happen. I think students sometimes feel pressured to compete, so they mention skills they have only done on basic level. It may not always be intentional exaggeration, but rather than an attempt to appear more competitive.

3. How do you usually check or evaluate a student's practical or technical skills?

- ❖ Well, I usually evaluate skills through project work, lab performance, presentations, coding, assignments, and viva

4. Do you think teachers should help verify student skills for jobs? Why or why not?

- ❖ Yes, I believe teachers can play an important role in verification. Since we as teachers directly observe students' performance and projects, we are in a good position to assess their actual capabilities.

5. What should be considered when verifying a student's skills?

- ❖ Well, I believe verification should be based on clear criteria such as project quality, consistency of performance, practical implementation, and understanding of concepts

6. Do you have any concerns about verifying student skills on a digital platform?

- ❖ My main concerns would be fairness, data privacy, and ensuring that the verification process is standardized across departments also it should protect the student and teacher data from getting misused

7. Do you think verified skill badges would help recruiters trust students more?

- ❖ Yes, I believe verified badges could improve credibility and multiple engagement.

8. How could this platform help connect universities with companies?

- ❖ It could also help universities showcase the practical strengths of their students, creating stronger industry-academia collaboration.

9. How can we make sure the verification process is fair for all students?

- ❖ well for that there should be some criteria or definitely transparencies in decision and choosing also you can add review mechanism sort of to give guidelines.

10. What features would you suggest to improve this platform?

- ❖ I would suggest that you can add evaluation forms or evidence should be uploaded work analytics etc.

Students:

Interviewee 02:

Name : Aunjolee Nusrath Anu

Age : 23 years old

Occupation: Student (University Of Asia Pacific)

Interviewer: MD. Fairuz Anam

I selected her for the interview because she provides a valuable and realistic viewpoint on the difficulties students encounter in the academic setting.

She is aware of the typical challenges concerning classes, instructors, assessments, amenities, and the communication divide between students and the administration.

Her perspectives on academic challenges, feedback issues, and the necessity for a clearer and more accessible system position her as a highly pertinent candidate for addressing and enhancing the System

1. Do you think there is a proper platform only for students to showcase their skills?

- ❖ To be honest, I don't think there is a proper centralized platform just for students to showcase their skills in a structured or verified way. I mostly depend on resumes or LinkedIn, but nothing is fully organized or official.

2. What problems do you face when applying for internships or jobs?

- ❖ A major problem that I face is proving my skills. Even if I have completed projects or courses, it's hard to show real evidence in a clear way. Sometimes I feel like my resume doesn't fully represent what I can actually do, what I cannot that is why sometimes the person Infront doesn't really understand if I am capable or not.

3. Do you think your CV properly shows your real skills?

- ❖ No not completely. A CV is very limited in space, so I can only mention things briefly. It doesn't show my actual understanding or effort behind each skill.

4. Have you ever felt that recruiters did not trust your listed skills?

- ❖ Yes, I feel that recruiters sometimes doubt what we write, especially when many students list similar skills. Without proof, it's hard for them to know who is genuine or who is faking theirs.

5. Do you think some students exaggerate their skills on CVs?

- ❖ Yes, I think some students do exaggerate. Sometimes people list skills they only learned at a basic level. which makes it harder for genuine students to stand out.

6. Would you like your skills to be officially verified by a teacher? Why or why not?

- ❖ Yes, why not I would like that. If a teacher verifies my skills based on my projects or performance, it would feel more authentic and trustworthy.

7. Do you think verified skills would increase your chances of getting selected?

- ❖ Yes, why not If my skills are verified, recruiters might trust my profile more and take me more seriously.

8. Would you use a student-only skill platform for internships and jobs?

- ❖ Yes, I would use it but it would be fine for me if other people also have their say in here like teachers, recruiters, company people.

9. What features would you want in such a platform?

- ❖ I would want skill tracking, certificate uploads, project links, teacher verification, progress tracking, and maybe career guidance suggestions.

10. Do you think this type of platform could improve student employability?

- ❖ Yes, I think it could really help students become more confident and better prepared for jobs.

11. What concerns would you have about this system?

- ❖ I would be concerned about privacy and integrity whether the verification process would be fair for all students or not.

12. What suggestions do you have to improve this idea?

- ❖ The platform should be simple to use, fair and workflow should be structured and it have to be easy to use web interface.

Interviewee 03:

Name : Dola Zaman

Age : 23 years old

Occupation: Student of University of Canberra, Australia

Interviewer: MD. Fairuz Anam

I chose **Dola Zaman**, a student of Cyber Security at the University of Canberra, Australia, because of her strong enthusiasm for technology and her deep interest in cyber security and digital systems. Her academic focus reflects not only technical knowledge in areas such as network security, system protection, and modern computing technologies, but also a strong curiosity for understanding and solving complex security challenges. Her dedication to applying theoretical knowledge to practical solutions makes her a highly inspiring and suitable individual to interview.

1. Do you think there is a proper platform only for students to showcase their skills?

- ❖ Honestly, no. Most platforms like LinkedIn or job portals are for everyone, not specifically for students. There isn't a proper structured platform where students can showcase their skills in a verified and organized way.

2. What problems do you face when applying for internships or jobs?

- ❖ One big problem is lack of experience. Even if I have done projects or courses, companies usually ask for real job experience. Also, it is hard to prove my skills properly in a short CV.

3. Do you think your CV properly shows your real skills?

- ❖ Not fully. A CV is limited. I can only briefly mention what I know, but it does not show how deeply I understand something or how much effort I put into learning it.

4. Have you ever felt that recruiters did not trust your listed skills?

- ❖ Yes, sometimes. Especially when many students list similar skills, recruiters might doubt whether we actually know them well.

5. Do you think some students exaggerate their skills on CVs?

- ❖ Yes, I think some do. Sometimes students mention skills they only practiced once or learned at a very basic level. That makes it harder for genuine students to stand out.

6. Would you like your skills to be officially verified by a teacher? Why or why not?

- ❖ Yes, I would like that. If a teacher verifies my skills based on my projects or performance, it would make my profile more trustworthy and professional.

7. Do you think verified skills would increase your chances of getting selected?

- ❖ Definitely. Verified skills would give me more confidence, and recruiters might take my application more seriously.

8. Would you use a student-only skill platform for internships and jobs?

- ❖ Yes, I would. If it is easy to use and widely accepted, I would prefer it over general job portals.

9. What features would you want in such a platform?

- ❖ I would want skill verification, certificate uploads, project links, internship matching, progress tracking, and maybe career guidance suggestions.

10. Do you think this type of platform could improve student employability?

- ❖ Yes, I believe it could. It would help students present themselves better and connect directly with recruiters.

11. What concerns would you have about this system?

- ❖ I would be concerned about privacy and fairness. The verification process should be equal for all students, and personal data must be protected.

12. What suggestions do you have to improve this idea?

- ❖ The platform should be easy, fair, and transparent. The verification criteria should be clear, and the interface should be easy to use. Also, teachers should not be overloaded with too many requests.

Recruiter:

Interviewee 04:

Name : Ashiqur Rahman Bhuiyan

Occupation: Chief Operating Officer, Nitol Motor Limited

Interviewer: MD. Fairuz Anam

I chose **Mr. Ashiqur Rahman Bhuiyan**, Chief Operating Officer of Nitol Motor Limited, for this interview because of his extensive corporate experience and leadership in a leading automotive and business organization. His strategic decision-making role and direct involvement in recruitment and operational management provide valuable industry insight into hiring practices and talent evaluation. His professional experience in managing teams and assessing workforce capability makes him a highly relevant and knowledgeable individual to interview for understanding the expectations of recruiters and employers.

1. What challenges do you face when hiring students or fresh graduates?

- ❖ The biggest challenge is assessing real capability. Many students have similar academic backgrounds, but it is difficult to understand who can actually perform in a real work environment. Lack of practical experience is also a common issue.

2. Do you find it difficult to verify the skills mentioned in student CVs?

- ❖ Yes, very often. Students list many technical skills, but there is usually no clear proof of their proficiency level. It takes time to verify through interviews or tests.

3. How do you currently assess whether a student actually has the skills they claim?

- ❖ We usually conduct technical interviews, practical tests, or assignment-based assessments. Sometimes we review project portfolios or GitHub profiles. But this process takes extra time and resources.

4. Would you prefer candidates with verified skills from university teachers? Why?

- ❖ Yes, I would prefer that. If a skill is verified by a teacher who directly supervised the student's project or coursework, it adds credibility and reduces uncertainty in the shortlisting stage.

5. Do you think a student-only recruitment platform would be useful for your organization?

- ❖ Yes, it could be useful, especially for internship hiring. A focused platform would save time and reduce irrelevant applications.

6. What information is most important to you when shortlisting student candidates?

- ❖ Verified technical skills, relevant projects, problem-solving ability, communication skills, and consistency in academic performance. I also look at practical exposure rather than just CGPA.

7. Would skill-based filtering (e.g., verified programming, design, communication skills) help you hire faster?

- ❖ Absolutely. If I can filter candidates by verified skills and proficiency level, it would significantly reduce screening time.

8. Do you think a verified badge system would increase your trust in applicants?

- ❖ Yes, provided the verification process is reliable and standardized. A verified badge would immediately highlight serious and capable candidates.

9. What concerns would you have about using a university-based student recruitment platform?

- ❖ My main concerns would be fairness in verification, possible bias, and data privacy. Also, the platform must maintain accuracy and not allow manipulation of information.

10. What features would you suggest to make such a platform valuable for recruiters?

- ❖ Advanced filtering options, verified skill levels, portfolio integration, internship tracking history, application management tools, and a simple dashboard for shortlisting. Analytics on candidate performance would also be helpful.

Feasibility Report

A feasibility study helps to determine whether the Student Skills Trackers System that we are proposing is practical, affordable and useful for the Stakeholders. The system's feasibility is determined by following aspects.

1. Technical Feasibility

The project is technically achievable using modern web technologies.

- The system can be developed using Django / Laravel / Node.js for backend.
- Frontend can be built using HTML, CSS, Bootstrap, or React.
- A relational database (MySQL / PostgreSQL) can store student profiles and skill records.
- Role-based authentication can be implemented for Students, Teachers, Recruiters, and Admin.
- Skill verification workflow can be built using approval request modules.
- Filtering and search system can be implemented using database queries.
- Cloud hosting (AWS / DigitalOcean / Shared Hosting) can support deployment.
- No special hardware is required.

Conclusion: The project is technically possible because it can be built using existing web technologies like Django or other frameworks. The required tools and technologies are available and practical.

2. Economic Feasibility

The project is financially manageable and cost-effective.

- Development can be done using open-source software.
- No licensing or software purchase cost.
- Hosting cost is minimal and affordable.
- Maintenance cost is low.
- No physical infrastructure required.
- Revenue model can include recruiter subscription or premium features.
- Universities may adopt the system officially in future.

Conclusion: The project is economically viable and sustainable. The system can be developed using open-source tools, so there is no high software cost. Hosting cost is low.

Professionally Developed System (Full Cost Estimation)

Item	Estimated Cost (BDT)
Backend Developer (3–4 months part-time)	90,000 – 120,000
Frontend Developer (3–4 months part-time)	60,000 – 80,000
UI/UX Design	20,000 – 30,000
Quality Assurance / Testing	15,000 – 25,000
Project Management (Optional)	20,000 – 30,000
Domain Name (1 year)	2,000
Cloud Hosting / VPS (1 year)	18,000 – 25,000
SSL Certificate	0 – 5,000
Database Setup & Configuration	5,000
Security & Backup Setup	10,000
Maintenance (Annual)	25,000 – 40,000
Miscellaneous / Contingency (10%)	30,000 – 40,000

3. Operational Feasibility

The system is practical and usable within university operations.

- Students can easily create and update profiles.
- Teachers can verify skills through a simple dashboard.
- Recruiters can search and filter students efficiently.
- User-friendly interface ensures smooth operation.
- University email authentication can ensure real student access.
- Reduces paperwork and manual verification.

Conclusion: The system can function effectively in real university environments.

4. Schedule Feasibility

The project can be completed within an academic semester.

Development Phases:

- Requirement analysis
- System design
- Database design
- Frontend development

- Backend development
- Skill verification module integration
- Testing and debugging
- Final deployment

Conclusion: The timeline is realistic and achievable. The project can be completed within one semester if development is properly planned. It does not require extremely complex infrastructure.

5. Legal & Ethical Feasibility

The project follows ethical and legal standards.

- Student data will be stored securely.
- Only verified and approved information will be visible.
- Students will have control over profile visibility.
- Teacher verification prevents false claims.
- Data privacy policies can be implemented.

Conclusion: The system promotes transparency and ethical recruitment.

6. Social Feasibility

The project has positive social and educational impact.

- Improves student employability.
- Builds trust between university and industry.
- Encourages honest skill representation.
- Reduces fake or exaggerated CV claims.
- Helps recruiters identify genuine talent.

Conclusion: The system benefits students, teachers, and recruiters.

Functional Requirements

1) User Registration and Login

- The system shall allow Students, Teachers, Recruiters, and Admin to register and log in.
- The system shall support role-based access (each role sees different features).
- The system shall allow password reset and account recovery.

2) Student Profile Management

- The system shall allow students to create, view, edit their profiles.
- The system shall store student information such as:
 - Basic info (name, contact, university, department, semester)
 - Academic info (CGPA, courses, achievements)
 - Non-academic info (clubs, volunteering, competitions)

3) Skill Management

- The system shall allow students to add, edit, delete skills (academic + non-academic).
- The system shall categorize skills (e.g., Technical, Soft skills, Creative, Leadership).
- The system shall allow students to attach proof (certificate, link, project, portfolio).

4) Teacher Skill Verification

- The system shall allow students to request verification for selected skills.
- The system shall notify the selected teacher/supervisor about requests.
- The system shall allow teachers to approve, reject, or request changes.
- The system shall allow teachers to assign a proficiency level (Beginner/Intermediate/Advanced) and add remarks.
- The system shall display a verified badge/status on approved skills.

5) Portfolio and Project Showcase

- The system shall allow students to add projects, descriptions, and links (GitHub, Drive).
- The system shall allow uploading or linking certificates and documents.

6) Recruiter Search and Filtering

- The system shall allow recruiters to search students by:
 - Skill (verified only / all)
 - Skill level
 - University, department, semester
 - CGPA range
 - Experience/projects
- The system shall allow recruiters to view student profiles with verified information clearly marked.

7) Job/Internship Posting and Applications

- The system shall allow recruiters to post internships/jobs with requirements.
- The system shall allow students to browse and apply.
- The system shall allow recruiters to shortlist/reject applicants.

8) Communication and Notifications

- The system shall send notifications for:
 - Verification requests and results
 - Job posts matching skills
 - Application status updates
- The system shall support recruiter-to-student messaging or contact request (optional but useful).

9) Admin Management

- The system shall allow admin to:
- Manage users (approve/disable accounts)
- Manage universities/departments/skill categories
- Monitor verification activities
- Handle reports/complaints (fake accounts, misuse)

10) Reporting and Analytics

- The system shall generate reports such as:
- Number of verified skills per student/department
- Most demanded skills by recruiters
- Placement/internship statistics (if tracked)

11) Security and Data Control Functions

- The system shall allow students to control profile visibility (public to recruiters / limited).
- The system shall keep logs of important actions (verification, admin actions).

12) University Email Authentication

- The system shall allow student registration using official university email addresses.
- The system shall verify email authenticity before account activation.
- The system shall restrict duplicate accounts using the same university ID.

13) Skill Endorsement System

- The system shall allow teachers to endorse specific student skills without full verification.
- The system shall display endorsement count on student profiles.

14) Skill Rating & Ranking

- The system shall allow proficiency levels (Beginner, Intermediate, Advanced, Expert).
- The system shall calculate a skill score based on verification and endorsements.

15) Internship/Job Matching System

- The system shall automatically match students with relevant job/internship posts based on skills.
- The system shall recommend jobs to students based on verified skills.

16) Application Tracking System

- The system shall allow students to track application status (Applied, Shortlisted, Rejected, Selected).
- The system shall allow recruiters to update application progress.

17) Profile Visibility & Privacy Settings

- The system shall allow students to hide specific skills or personal information.
- The system shall allow students to make profiles public or private to recruiters.

18) Feedback & Review System

- The system shall allow recruiters to provide feedback after internship completion.
- The system shall allow students to receive performance remarks.

19) Search History & Saved Candidates

- The system shall allow recruiters to save shortlisted candidates.
- The system shall allow recruiters to view previous search history.

20) System Audit & Activity Monitoring

- The system shall track verification requests and approval history.
- The system shall log system activities for transparency and accountability.

Non-Functional Requirements

1. Performance Requirements

- The system shall support at least 500–1000 concurrent users without performance degradation.
- Profile loading time shall not exceed 2 seconds under normal conditions.
- Skill verification updates shall reflect in the system within 3 seconds.
- Search and filtering results shall generate within 5 seconds.

2. Security Requirements

- The system shall implement role-based access control (Student, Teacher, Recruiter, Admin).
- All passwords shall be securely encrypted.
- Data transmission shall use secure communication protocols (HTTPS/TLS).
- The system shall prevent unauthorized access to student data.
- Only verified teachers shall be allowed to approve skills.
- The system shall maintain activity logs for important actions.

3. Usability Requirements

- The interface shall be simple and user-friendly.
- The system shall be accessible via desktop and mobile browsers.
- Navigation shall be clear and easy to understand.
- Forms shall provide validation messages and guidance.
- New users shall be able to complete profile setup within 10 minutes.

4. Reliability Requirements

- The system shall have at least 99% availability.
- The system shall automatically save data to prevent data loss.

- Backup of database shall be performed regularly.
- The system shall handle errors gracefully without crashing.

5. Scalability Requirements

- The system shall support increasing numbers of users without major redesign.
- The architecture shall allow adding new universities or departments easily.
- The system shall allow future integration with university systems or APIs.

6. Maintainability Requirements

- The system shall use modular and clean code structure.
- Documentation shall be maintained for future updates.
- Updates and bug fixes shall be possible without affecting core operations.

7. Compatibility Requirements

- The system shall support major web browsers (Chrome, Edge, Firefox).
- The system shall be responsive across different screen sizes.

8. Legal & Privacy Requirements

- Student data shall be stored securely.
- Students shall have control over profile visibility.
- The system shall comply with data protection and privacy standards.
- Personal data shall not be shared without user consent.

9. Availability & Disaster Recovery Requirements

- The system shall provide continuous service with minimal downtime.
- The system shall have an automatic backup and disaster recovery mechanism.
- The system shall restore data within an acceptable recovery time in case of system failure.

10. Accessibility Requirements

- The system shall be accessible to users with basic digital literacy.

- The system shall use readable fonts, proper contrast, and clear interface design.
- The system shall support accessibility standards to ensure usability for all students.

11. Audit & Monitoring Requirements

- The system shall record system logs for login attempts and verification activities.
- The system shall allow administrators to monitor suspicious activities.
- The system shall provide traceability for verification approvals and modifications

Conclusion

The **Student Skill Tracker Platform** addresses a critical and growing need in modern education by providing a structured, transparent, and reliable framework for skill development and verification. By combining student self-assessment, instructor validation, and data-driven insights, the platform supports outcome-based education and significantly enhances career readiness. Its technical depth, multi-stakeholder involvement, and real-world impact clearly qualify it as a **Complex Engineering Problem**.

